WORKING PAPER

MASS Regulatory Bridge

Autonomous Ships and the Legacy Instruments

Interface Issues between COLREG, SOLAS, MARPOL, STCW/ISM/ISPS and the MASS Code (MSC.595(111))

Working paper for discussion. Analytical guidance only; not an IMO instrument.

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1. Executive Summary

Purpose. This paper identifies where the legacy IMO instruments (COLREG, SOLAS, MARPOL, STCW, the ISM and ISPS Codes) and the MASS Code (MSC.595(111)) interact awkwardly, sets out what each friction means in practice, and recommends how a submitter, an Administration or a class society can deal with it in the approval basis, the concept of operations (ConOps) and the design of the autonomous navigation system (ANS). It is written for readers who already work with these instruments.

Key findings.

1. **The mandatory baseline has not moved.** The MASS Code is non-mandatory. Until it is made mandatory (the roadmap target is 1 January 2032, through a new SOLAS chapter XVI), SOLAS, COLREG, MARPOL, STCW, ISM and ISPS remain the binding requirements, together with flag, coastal and port State law. Equivalence arguments under the Code are made by the submitter and accepted by the Administration. They do not amend or relax the legacy text.
2. **Most friction sits where legacy rules use human judgment terms.** "Look-out by sight and hearing", "safe speed", "scanty information", "readily apparent", "as soon as it becomes apparent" and "immediate danger" have no numeric content. The Code answers in functional terms (situational awareness, operational envelope, human oversight) but sets no thresholds. The thresholds have to be declared and justified vessel by vessel.
3. **Responsibility stays with a human, but the mechanics are unspecified.** The Code keeps a human master responsible at all times, and the ANS is not a master. It does not set a maximum intervention latency, does not say who reports incidents under MARPOL, and does not allocate navigation-safety responsibility across flag State, Company, remote operation centre (ROC) and master. These must be settled in the ConOps.
4. **Environmental obligations are the least developed area.** The Code has no MARPOL annex-specific treatment. Annexes I and VI are the most open. Annexes IV and V cannot be dismissed on the basis that there is no crew.
5. **Ever Smart ([2021] UKSC 6) gives two directly usable tests.** Crossing rules engage on a steady *bearing*, not a steady *course*. Narrow-channel rules displace the crossing rules at a channel entrance only once the approaching vessel is *shaping to enter* on final approach, not merely intending or preparing to enter. These are UK law and persuasive elsewhere, and both bear directly on ANS design.
6. **During the voluntary period the approval basis, ConOps and regulatory gap analysis are the working interpretation layer.** They are where a vessel-specific reading of an unchanged COLREG is recorded, justified and made a condition of approval. COLREG is unlikely to be amended before 2032, so this layer carries most of the weight in the interim.
7. **Verification operates at several distinct levels** (flag-State audit, COLREG Part F, casualty investigation, vessel and ROC certification). They should be kept separate and none treated as a substitute for another.
8. **Much of the detailed design guidance in this paper is proposal, not settled practice.** The rule-switching state machine in Section 5.4 and its numeric parameters are engineering proposals that no flag State, class society or IMO body has validated.

How to use this paper. Section 3 can be used as a checklist when preparing or reviewing a regulatory gap analysis. Section 4 should be read before encoding any collision-avoidance rule logic. Section 5 is a drafting guide for the ConOps and approval basis. Section 7 lists what must be verified before anyone relies on this paper for certification or legal purposes.

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2. Scope, Authority and Limitations

2.1 What this paper is and is not

It is a technical and regulatory analysis intended to support approval work and design decisions. It is not an IMO interpretation, not legal advice, and not a statement of any Administration's position. Recommendations use "should" to describe suggested practice. Nothing here creates an obligation.

2.2 Status of the MASS Code

- MSC.595(111) was adopted on 21 May 2026 and invited Governments to give effect to it from 1 July 2026. It is expressly non-mandatory (MASS Code 1.2.4).
- The roadmap annexed to the resolution expects a mandatory Code to enter into force on 1 January 2032 through SOLAS chapter XVI. That date is a target, not current law.
- The Code states that it supplements other applicable instruments (1.2.1) and that its evaluation criterion is a level of safety, security and environmental protection expected of a conventional ship (6.2).
- **Reading convention used in this paper.** Where the paper says the Code "requires" something, read it as "requires, as guidance, unless made binding by national implementation".

2.3 Application of the Code

- It applies to cargo ships to which SOLAS chapter I applies, including associated ROCs, where the Administration deems compliance with other applicable instruments impracticable or insufficient (2.1). The application decision therefore rests with the Administration and should be recorded (Section 3.5.3).
- It excludes cargo high-speed craft to which SOLAS chapter X applies, warships, naval auxiliaries and other government non-commercial ships.
- For MASS below 500 gross tonnage, the resolution invites application "as far as practicable". No mandatory small-MASS regime exists.

2.4 Mandatory legacy instruments and the voluntary Code

Layer	Status in 2026–2032	Role in this paper
SOLAS, COLREG, MARPOL, STCW, ISM, ISPS	Binding on Parties as implemented	Source of the obligations whose application to MASS is analysed
Flag, coastal and port State law	Binding within jurisdiction	Determines which interpretations apply
MASS Code MSC.595(111)	Voluntary guidance unless adopted nationally or by contract or class requirement	Framework for demonstrating equivalence and structuring approval
Approval basis, ConOps, regulatory gap analysis	Binding on the vessel and Company to the extent the Administration makes them conditions of approval	Vessel-specific interpretation layer (Section 5.2)

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2.5 Sources and their limits

- The MASS Code text and the IMO Status of Treaties volume (30 October 2026) were the primary sources for the earlier drafts. The Status of Treaties volume gives amendment histories and final clauses. It does not reproduce the operative text of SOLAS regulations, COLREG rules or MARPOL annexes.
- Operative COLREG wording used here is taken from text quoted in the *Ever Smart* judgment and from a secondary technical source. It has not been checked against the consolidated COLREG text.
- References to SOLAS, STCW, Load Lines and MARPOL operative provisions are made at the level of general content and should be verified (Section 7).
- Flag-State, class-society and port-State practice on MASS has not been surveyed for this paper.

3. Core Friction Analysis

3.1 Approach

Each major friction is described under five headings: the legacy requirement, how the MASS Code treats it, the nature of the friction, the practical implications, and the recommended approach. Minor items are tabulated.

The Code's evaluation principle (6.2) frames every item. For each friction the working question is not whether the MASS satisfies the literal wording by the same means a crewed ship would, but whether the submitter can show a level of safety, security and environmental protection at least equal to that of a conventional ship. That is an argument made in the approval process and accepted, or not, by the Administration. It does not change the legacy text.

3.2 Summary of frictions

Ref	Issue	Principal instrument	Position
3.3.1	Look-out by sight and hearing	COLREG 5	Addressed in principle; equivalence shown per vessel
3.3.2	Scanty information	COLREG 7	No threshold; must be declared
3.3.3	Restricted visibility	COLREG 19	Partly addressed; Rule 19 specifics not encoded
3.4.1	Safe speed against operational envelope	COLREG 6	Partly addressed; boundary behaves as a cliff
3.4.2	Readily apparent action and its timing	COLREG 8	Partly addressed
3.4.3	Stand-on vessel triggers	COLREG 17	No thresholds; observation and prediction problems differ
3.4.4	Crossing-rule engagement	COLREG 7(d)(i), 15	See Section 4
3.4.5	Lights, shapes, sound and light signals	COLREG Parts C, D	Applicability clear; activation and verification open
3.4.6	Non-cooperative targets	COLREG, Code 16.4.3	Not addressed
3.5.1	Narrow channels, traffic separation, small MASS	COLREG 9, 10	Open for small MASS
3.5.2	Rule 9 against crossing rules	COLREG, case law	See Section 4
3.5.3	Rule sets by geography	COLREG, national rules	Open; must be declared

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Ref	Issue	Principal instrument	Position
3.6.1	Remote master and departures from the Rules	COLREG 2	Latency undefined
3.6.2	Limitations of the vessel and crew	COLREG 2	Substitution by human-system analysis
3.6.3	Allocation of responsibility	SOLAS, COLREG, ISM	Open at IMO level; ConOps matrix
3.6.4	Safety management and the ROC	ISM Code	Alternative certification available
3.6.5	Qualification, watchkeeping and rest	STCW	Partly addressed
3.7.1	Verification layers	SOLAS XI-1/6, COLREG Part F, III Code, Code Ch. 5–6	Layered; keep distinct
3.7.2	Navigational equipment and VDR	SOLAS V/19, V/20	Partly addressed
3.7.3	Application gate and amendment drift	Code 2.1; all instruments	Open; record and version-lock
3.8	Stability, fire, life-saving, Load Lines	SOLAS II-1, II-2, III, V/33; Load Lines	Mostly addressed by automation or modification
3.9.1	Oil pollution	MARPOL Annex I	Code gap
3.9.2	Sewage, garbage, air, incident reports	MARPOL Annexes IV, V, VI; Protocol I	Conditional or open
3.9.3	Security and cyber	SOLAS XI-2, ISPS	Addressed by extension; link to COLREG functions missing

3.3 Look-out and Situational Awareness

3.3.1 Look-out by sight and hearing (COLREG Rule 5)

- **Legacy requirement.** A proper look-out by sight and hearing, as well as by all available means appropriate in the prevailing circumstances and conditions.
- **MASS Code treatment.** The ANS should maintain adequate situational awareness, with continuous monitoring of information necessary for safe navigation within the operational envelope (16.4.2). Situational awareness is defined as perception, comprehension and projection (4.33).
- **Nature of the friction.** Rule 5 names human senses. An ANS uses sensor fusion (radar, electro-optical and infrared cameras, AIS, LIDAR, acoustic sensing). The Code supplies a functional standard, but the Rule text is unchanged and sets no minimum.
- **Practical implications.** Compliance has to be demonstrated as equivalence for each vessel. There is no type-neutral acceptance criterion, so two Administrations may reach different conclusions on the same sensor suite.
- **Recommended approach.** Declare the sensor suite, coverage, degradation modes and the minimum configuration under which ANS control may continue. Map the suite to the Rule's functions (optical and electro-optical sensing to sight, acoustic sensing to hearing, AIS, VHF and data links to other available means). Record the equivalence argument in the regulatory gap analysis. Do not describe the Rule as amended.

3.3.2 Assumptions on scanty information (COLREG Rule 7)

- **Legacy requirement.** No assumptions are to be made on the basis of scanty information, especially scanty radar information.
- **MASS Code treatment.** The ANS continuously analyses and projects future status (16.4.2); avoidance action is based on risk assessment (16.4.3); alerts go to the master (13.4.1.4).

- **Nature of the friction.** An ANS may act on degraded or incomplete sensor data. "Scanty" has no data-quality threshold anywhere in the Code.
- **Practical implications.** As written the prohibition cannot be tested. An approval team will reasonably ask for a measurable criterion.
- **Recommended approach.** Define a sensor sufficiency and confidence floor by encounter type and operational envelope. Below the floor the ANS should alert the master and enter a defined degraded state. Show that the ANS's handling of uncertainty is at least as conservative as that of a prudent officer. Whether IMO will ever set a common threshold is unknown.

3.3.3 Restricted visibility (COLREG Rule 19)

- **Legacy requirement.** Safe speed adapted to the circumstances, and a duty on a vessel that detects another by radar alone to determine whether a close-quarters situation is developing. Rule 19 also limits certain avoiding alterations of course (for example, alterations to port for a vessel forward of the beam). The detail should be confirmed against the consolidated text.
- **MASS Code treatment.** Situational awareness is sensor-agnostic (16.4.2, 16.4.3). Visibility limits belong in the operational envelope (8.3.1.2.3).
- **Nature of the friction.** Rule 19 assumes radar as the primary sensor. A fused system has no direct analogue of "detects by radar alone", and the Code does not encode Rule 19's specific manoeuvring restrictions.
- **Practical implications.** Without a declared restricted-visibility mode, the ANS's behaviour in fog cannot be assessed against the Rule.
- **Recommended approach.** Declare the restricted-visibility mode, how visibility is determined from sensor data, the speed policy and the manoeuvres the ANS will not make. Define the fused-sensor condition treated as equivalent to detection by radar alone (for example, a target held on radar without corroboration from other sensors). Link to the automated sound signals discussed at 3.4.5.

3.4 Collision Avoidance

3.4.1 Safe speed (COLREG Rule 6)

- **Legacy requirement.** Safe speed at all times, taking into account visibility, traffic density, manoeuvrability, background light, wind, sea and current, draught and radar characteristics.
- **MASS Code treatment.** The operational envelope includes environmental limitations (8.3.1); the ANS maintains functionality under expected conditions (16.3.2); transition to degraded or fallback states is governed by 8.3.3 and 8.4.
- **Nature of the friction.** Rule 6 is a continuous judgment. The operational envelope is a discrete boundary, so leaving it behaves as a cliff rather than a gradient.
- **Practical implications.** An ANS that is fully capable at the boundary and abruptly degraded beyond it will either operate at the edge of its capability or hand over at the worst moment.
- **Recommended approach.** Convert each Rule 6 factor into a monitored envelope parameter with a margin. Treat safe speed as the speed at which the MASS stays inside its envelope with margin. Define degraded-state buffers before the envelope limit, with graduated speed and mode changes, so that the boundary is approached progressively.

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3.4.2 Action that is readily apparent, and when it is taken (COLREG Rule 8)

- **Legacy requirement.** Action to avoid collision is to be positive, made in ample time and large enough to be readily apparent to another vessel observing visually or by radar.
- **MASS Code treatment.** The ANS has the means to act in accordance with COLREG, and decisions are timely and within safe operating limits (16.4.3).
- **Nature of the friction.** The other vessel may be crewed (human look-out) or a MASS (sensor-based). Small incremental manoeuvres may be unreadable to either. The Code does not say how ANS action is made apparent to others. The UK Supreme Court (para 40, citing *The Maloja II*) also stresses timing: manoeuvres to avoid a close-quarters situation should be made when the officer is not forced into a quick decision or a decision on inadequate information, and should be readily apparent to the other ship.
- **Practical implications.** "Readily apparent" is as much a question of *when* the ANS decides as of how large the manoeuvre is. An ANS that waits too long will fail the test whatever the size of its eventual manoeuvre.
- **Recommended approach.** Set the decision threshold early enough that (1) the decision is not made under time pressure, (2) it is not made on inadequate information and (3) the resulting manoeuvre is large enough to be readily apparent. Declare a maximum decision latency in the ConOps. Validate legibility against both human visual and radar observers and other MASS sensors. Digital signalling (AIS, VDES) may supplement MASS-to-MASS encounters but does not replace the Rule's standard for crewed vessels.

3.4.3 Stand-on vessel (COLREG Rule 17)

- **Legacy requirement.** The stand-on vessel may act by her manoeuvre alone as soon as it becomes apparent that the give-way vessel is not taking appropriate action (17(a)(ii)). When collision cannot be avoided by the give-way vessel's action alone, she shall take the action that best aids avoidance (17(b)).
- **MASS Code treatment.** Collision avoidance (16.4.3); situational awareness including projection of future status (4.33, 16.4.2); human ability to intervene and override (8.7.1, 16.5); V&V plan (Annex 1, 2.7).
- **Nature of the friction.** Both sub-rules rest on human judgment, and they raise different technical problems. The 17(a)(ii) trigger is an *observation* problem: has the give-way vessel's behaviour departed from expected compliant behaviour beyond a declared threshold? The 17(b) trigger is a *prediction* problem: can the give-way vessel avoid collision alone? The judgment treats 17(a)(ii) as a permissive, observation-based threshold and 17(b) as a mandatory, prediction-based one (paras 60–61). The Code requires projection but sets no accuracy requirement.
- **Practical implications.** Different ANS designs will set different thresholds, and neither threshold is bounded by any standard. Prediction requirements are the stricter of the two, and this is probably the hardest part of COLREG to implement.
- **Recommended approach.** Declare both thresholds separately in the approval basis. For 17(a)(ii), declare a CPA/TCPA or equivalent risk threshold and the change-detection parameters. For 17(b), declare prediction-accuracy requirements, false-negative and false-positive limits and escalation to the master when accuracy falls below the declared level. Validate against encounter scenarios, including erratic give-way behaviour. Preserve override under 16.5.

3.4.4 Crossing-rule engagement (COLREG Rules 7(d)(i) and 15)

The UK Supreme Court's holding that the crossing rules engage on steady bearing rather than steady course changes the minimum sensing and logic requirements of an ANS. It is treated in Section 4.4.

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3.4.5 Lights, shapes, sound and light signals (COLREG Parts C and D)

- **Legacy requirement.** Rules 20–31 (lights and shapes) and Rules 32–37 (sound and light signals), including Rule 35 in or near restricted visibility.
- **MASS Code treatment.** COLREG compliance is required (16.2). The MASS Safety Certificate and MASS Record are issued in addition to the SOLAS chapter I certificates (5.1.1, 5.1.5). There is no dedicated chapter on lights, shapes or sound signals.
- **Nature of the friction.** Applicability is not in doubt. The gap is practical: what activates sound signals when no one is on the bridge, and how Parts C and D are verified at survey for an unmanned ship.
- **Practical implications.** A safety-critical signalling duty may be compliant in principle and untested in practice.
- **Recommended approach.** Add a survey and V&V checklist covering lights and shapes, automatic Rule 35 signals triggered by the restricted-visibility mode, fault monitoring, and manual or remote override. State in the ConOps who or what initiates signals in each mode.

3.4.6 Non-cooperative targets

- **Legacy requirement.** None specific; COLREG assumes vessels that comply or can be assumed to try to.
- **MASS Code treatment.** Risk-based collision avoidance (16.4.3). Performance against AIS-disabled, deceptive or non-compliant vessels is not addressed.
- **Nature of the friction and implications.** Real encounters include vessels that do not behave as the Rules expect. The Code gives no basis for judging ANS performance against them.
- **Recommended approach.** Include non-cooperative target scenarios in V&V. Declare in the ConOps the capability limits and the escalation to the master or ROC when those limits are reached.

3.5 Narrow Channels and Traffic Separation

3.5.1 Narrow channels, traffic separation schemes and small MASS (COLREG Rules 9 and 10)

- **Legacy requirement.** Rule 9 governs vessels in narrow channels and fairways. Rule 10 governs traffic separation schemes, and includes a provision that a vessel under 20 metres shall not impede a power-driven vessel following a traffic lane (10(j); to be confirmed against the consolidated text).
- **MASS Code treatment.** The ANS executes the voyage plan within allowable deviation (16.4.4). Below 500 gross tonnage the Code is invited "as far as practicable" and no mandatory regime applies.
- **Nature of the friction.** Constrained-water behaviour must be encoded in the voyage plan and the ANS logic. A small MASS in a traffic separation scheme is bound by Rule 10 but may not be covered by the Code.
- **Practical implications.** Small-MASS operation in COLREG waters depends on national rules, and flag-State practice has not been surveyed.
- **Recommended approach.** Declare in the ConOps the rule set applied, including behaviour in traffic separation schemes and narrow channels. Flag States should consider defining small-MASS rules. Future IMO guidance for small MASS would be the long-term answer.

3.5.2 Rule 9 against the crossing rules at channel entrances

See Section 4.3 and the proposed design approach in Section 5.4.

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3.5.3 Rule sets by geography

- **Legacy requirement.** International COLREG at sea. National inland rules inside demarcation lines (for example, the U.S. Inland Navigation Rules, 33 CFR Part 83, whose rule numbering and content differ in places from the international Rules).
- **MASS Code treatment.** None specific.
- **Nature of the friction and implications.** A MASS passing a demarcation line changes the rules that govern it, and the ANS must change behaviour at the right point.
- **Recommended approach.** Declare the applicable rule set by voyage phase and geography in the ConOps, and test transitions in V&V. Verify the current text of the national rules before relying on this point.

3.6 Responsibility and Intervention

3.6.1 Remote master and departure from the Rules (COLREG Rule 2)

- **Legacy requirement.** Rule 2 assigns responsibility to vessel, owner, master and crew, and permits departure from the Rules to avoid immediate danger.
- **MASS Code treatment.** A human master is responsible for the safe operation of the MASS at all times (8.7.3, 14.2.1.10), and the master may be on board or at a ROC. If persons are on board, the master must be physically on board (14.2.1.12). Fallback responses are provided for (8.4.2).
- **Nature of the friction.** Rule 2 assumes a master on the bridge who can judge immediate danger. At a ROC that judgment is made at a distance and with latency, and the Code specifies no maximum latency.
- **Practical implications.** Between detection of danger and master action the ANS is acting alone, and what it may do in that interval is a design decision with legal consequences.
- **Recommended approach.** Declare the maximum latency from danger detection to master intervention for each operating phase, the ANS fallback response during that interval and the transfer of authority. Verify that the ANS's default fallback conforms to Rule 2 and the other Rules. State the condition for requiring the master on board.

3.6.2 Limitations of the vessel and her crew (COLREG Rule 2)

- **Legacy requirement.** Due regard to the limitations of the vessel and her crew.
- **MASS Code treatment.** Crew and remote operators are to be always able and prepared to take over functions without risk to the task (8.7.2, 14.2.1).
- **Nature of the friction.** With no crew on board the clause appears vacuous.
- **Recommended approach.** Substitute a human-system limitation analysis covering remote operators and the ANS, and record it in the regulatory gap analysis. The absence of crew does not mean the absence of limitations.

3.6.3 Allocation of responsibility

- **Legacy requirement.** Navigation-safety responsibility is spread across the master (COLREG Rule 2), the Company (ISM) and the flag State (SOLAS).
- **MASS Code treatment.** The Code preserves a human master, Company safety management, ROC management and Administration approval (8.6.2, 8.7, 11, 14, Annex 1, 2.4). It defines "agent" to include software (4.1) but reserves the master role for a human (8.7.3).
- **Nature of the friction.** The ANS is an agent that acts, while a human is responsible for what it does. The master may be at a ROC, the Company operates the ROC, and the flag State certifies both.

- **Practical implications.** The chain flag State, Company, ROC, master, ANS runs across several instruments. Gaps are likely at the joins, particularly intervention, reporting and fallback authority.
- **Recommended approach.** Prepare a responsibility-allocation matrix (Section 5.5) and use the approval process to confirm that no responsibility gap exists.

3.6.4 Safety management and the ROC (ISM Code)

- **Legacy requirement.** The ISM chain runs Company, master, crew, evidenced by a Document of Compliance and Safety Management Certificate.
- **MASS Code treatment.** ROC safety management system, Document of Compliance for the ROC management company, and ROC Management Certificate (5.7, 11.2, 11.3).
- **Nature of the friction.** The ROC is an additional node not accounted for by the standard certificates. The relationship becomes Company, MASS, ROC and master.
- **Recommended approach.** Use the SMS together with the ROC safety management system where alternative ROC management is approved, and define the responsibility interfaces explicitly.

3.6.5 Qualification, watchkeeping and rest (STCW)

- **Legacy requirement.** Qualification and sea-going service, onboard watchkeeping, and hours-of-rest requirements written for seafarers (A-VIII/1).
- **MASS Code treatment.** Remote operators are qualified to a level not less than STCW 1978 (14.2.2.1). A ROC may be treated as a location directly associated with the navigational bridge (14.1.3). Remote operating experience may be considered equivalent to approved sea service (14.5.1). Fatigue and rest principles apply (14.2.1.8–.9), and the ROC must have sufficient operators, time, resources and procedures (14.2.2.3).
- **Nature of the friction.** Fatigue principles exist. What is missing is ROC-specific quantification, such as rest-hour limits and the number of vessels one operator may supervise.
- **Recommended approach.** Adopt a ROC watchkeeping standard with rest-hour records and workload limits, and assess alarm load for operators supervising multiple MASS. Confirm the applicability of STCW provisions to remote operators with the Administration.

3.7 Verification and Certification

3.7.1 Verification layers

- **Legacy requirement.** COLREG Part F (flag-State audit under the IMO Member State Audit Scheme); the III Code (mandatory through SOLAS chapter XIII from 1 January 2016); SOLAS XI-1/6 (casualty investigation).
- **MASS Code treatment.** Chapters 5 and 6 add vessel and ROC surveys, certificates and an approval process.
- **Nature of the friction.** Four regimes verify at different levels of granularity: State, Rule, casualty and vessel.
- **Recommended approach.** Keep the layers distinct. Present the evidence stack (flag-State audit, MASS and ROC certificates, vessel incident data) without suggesting that one layer can discharge another. No strict hierarchy is asserted.

3.7.2 Navigational equipment and voyage data recorders (SOLAS V/19 and V/20)

- **Legacy requirement.** Carriage requirements for shipborne navigational systems (V/19) and voyage data recorders (V/20).
- **MASS Code treatment.** Certain V/19 requirements are to be met by electronic means (16.3.3 EP1), and V/20 applies to all MASS (9.10.6).

- **Nature of the friction.** V/19 assumes a human navigator operating equipment on a bridge. For a MASS the equipment must be machine-readable and remotely accessible.
- **Recommended approach.** Declare how each V/19 requirement is met electronically, including the ANS reading electronic chart data directly and the arrangements for redundant electronic charts. Verify the operative SOLAS text.

3.7.3 Application gate and amendment drift

- The Code applies only where the Administration deems other instruments impracticable or insufficient (2.1). The decision, the instruments judged insufficient and the chapters invoked should be recorded in the MASS Record and approval basis.
- Amendments adopted now will enter into force during the transition. According to the Status of Treaties volume, 2026 SOLAS amendments enter into force in 2028 and 2026 MARPOL Annex VI amendments in 2027. An approval basis proven against one rule version may be overtaken. The approval basis should be version-locked, and an amendment horizon matrix (adopted, accepted, in force, applicable to the vessel, version used in the approval basis) should be maintained.

3.8 Ship Systems: Stability, Fire, Life-Saving and Load Lines

Legacy provision	MASS Code treatment	Friction	Recommended approach
SOLAS II-1 (stability, watertight integrity)	Ch. 19; stability control system assesses intact stability and supports survivability assessment (19.2.2)	Criteria assume crew corrective action	Treat the stability control system as the monitoring and alerting function crew would have provided. Declare the human or remote response chain.
SOLAS II-2 (fire)	Ch. 20; remote and onboard control of active fire protection and interface for external responders (20.2.7, 20.2.8)	Firefighting assumes crew	Rely on automated detection, remote or autonomous extinction and an external-responder interface. Validate fire control and fallback logic.
SOLAS III and V/33 (life-saving; assistance to persons in distress)	Ch. 21; rescue equipment usable independently of crew; accommodation for retrieved persons (21.2.1–21.2.4)	Chapter III assumes crew, but duty to assist remains	Declare crewless deployment of rescue equipment, accommodation for rescued persons and communication with the ROC.
Load Lines Convention and 2008 IS Code	Compliance required (19.2.1)	Easily overlooked in MASS mapping	Include Load Lines and IS Code compliance in the certification mapping and verify operative text.

3.9 Environmental and Security Obligations

3.9.1 Oil pollution prevention (MARPOL Annex I)

- **Legacy requirement.** Oil discharge monitoring, oily water separation, IOPP certificate and surveys.
- **MASS Code treatment.** Environmental goals and general provisions (9.10, Ch. 24). No Annex I provision.
- **Nature of the friction.** Machinery spaces may be unattended, so monitoring and records must be automated. A MASS remains a ship under MARPOL, and the Code states that it is supplementary to other instruments (Preamble 5).
- **Practical implications.** How surveyors will treat unmanned machinery spaces under the IOPP regime is unsettled.
- **Recommended approach.** Annex I applies unchanged. Add an Annex I compliance checklist to the ConOps and MASS Record, automate discharge monitoring and logging, and confirm with the Administration how IOPP surveys will treat unmanned spaces.

3.9.2 Other MARPOL obligations

Provision	MASS Code treatment	Friction	Recommended approach
Annex IV (sewage)	21.2.4; 2.3 (modes with persons on board)	"No crew" does not mean no persons: maintenance crews, pilots, rescued persons and mixed modes	Apply whenever persons or sewage sources are present. Declare zero-person periods and holding or reception arrangements.
Annex V (garbage)	Ch. 22; 9.10; Ch. 24	Garbage arises from cargo residues, packaging, stores, maintenance and lost items regardless of crew	Add a garbage management plan by voyage phase and person-presence mode.
Annex VI (air pollution)	Ch. 24; machinery chapter covers safe and reliable machinery, not emissions (24.2.3–24.2.9)	Alternative propulsion (electric, hybrid, wind-assist) still has to meet emissions and fuel rules	Keep Annex VI compliance separate. Declare propulsion type and the compliance method or approved equivalent.
Protocol I (incident reports)	Logged data available to authorities on request (9.10.5); communications chapters (17, 18)	Data availability is not the same as an immediate report, and the Code does not say who reports	Assign the reporting duty to master, Company or ROC in the ConOps and SMS. Automate detection and log support. Verify communications availability.

3.9.3 Security and cyber (SOLAS XI-2 and the ISPS Code)

- **Legacy requirement.** Ship security measures including access control, restricted areas and an onboard ship security officer.
- **MASS Code treatment.** The ship security plan accounts for the ROC (12.4.1). Remote operators take part in ISPS training, drills and exercises (12.4.2.2). A ROC security breach triggers fallback (12.4.3). Cyber requirements appear in 9.8, 10.3, 12.3–12.6 and 17.2.10.
- **Nature of the friction.** The ROC becomes a new security target and there is no onboard security officer. Separately, a cyber event can degrade situational awareness or control and so cause navigational non-compliance (for example, a degraded look-out under Rule 5). The Code addresses cyber risk but does not link it to COLREG-critical functions. That link is an inference of this paper.
- **Recommended approach.** Treat the ROC as a security target in the ship security plan, with isolation and fallback on compromise. Do not assume the remote operator automatically replaces the ship security officer, and allocate transferred responsibilities explicitly. Link the cyber risk assessment to COLREG-critical functions and define cyber-induced degraded and fallback states.

4. Case Law Integration: Ever Smart and Rule Interaction

4.1 The case and its weight

Evergreen Marine (UK) Ltd v Nautical Challenge Ltd [2021] UKSC 6 (the *Ever Smart*) concerned a collision between two vessels at the entrance to a narrow channel. COLREG was not amended. The case settled how the narrow-channel and crossing rules interact by way of judicial interpretation.

The judgment is a decision of the UK Supreme Court. It binds as a matter of UK law and is persuasive elsewhere. It is not an IMO interpretation, and this paper has not surveyed which interpretation other flag or coastal States apply. Paragraph numbers below are carried from earlier analysis and should be re-checked against the judgment.

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4.2 Why it matters for autonomous systems

Both holdings supply tests that can be observed by sensors and implemented in logic. COLREG alone does not tell a system *when* to switch between rule sets, or what condition engages the crossing rules. For human mariners that is a matter of judgment. For an ANS it must be written down.

4.3 Rule 9 and the crossing rules at channel entrances (para 145)

The Court held, in substance, that the crossing rules are not displaced by the narrow-channel rules merely because an approaching vessel intends, or is preparing, to enter the channel. They are displaced only if and when that vessel is *shaping to enter*: adjusting her course so as to reach the entrance on her starboard side, on her final approach.

Read with the surrounding reasoning (paras 134–145), this gives three situations:

Group	Situation	Rules that apply
1	Vessel crossing the channel entrance, not intending to enter	Crossing rules
2	Vessel shaping to enter on final approach	Narrow-channel rule (Rule 9) overrides the crossing rules
3	Vessel intending or preparing to enter but not yet shaping, for example waiting for a pilot	Crossing rules still apply

The vessel in Group 3 in the case (ALEXANDRA 1) was waiting for a pilot, and the crossing rules were held to apply to her. A MASS leaving a channel while another vessel waits outside the entrance can face the same situation.

4.4 Bearing, not course (paras 111–112)

The Court held that where two vessels, both moving over the ground, are crossing so as to involve risk of collision, the engagement of the crossing rules does not depend on the give-way vessel being on a steady course, and the same applies to the stand-on vessel. The trigger is risk of collision shown in the way Rule 7(d)(i) describes it, namely a compass bearing of the approaching vessel that does not appreciably change.

For an ANS this is a design constraint. Logic that first asks whether the other vessel is on a steady course before applying the crossing rules will fail this standard. A vessel that is erratic, slow, manoeuvring or waiting remains subject to the crossing rules if the bearing is steady. The minimum sensing requirement is therefore accurate bearing measurement over time, not course estimation.

4.5 Rule 17 triggers (paras 60–61) and timing of manoeuvres (para 40)

- Rule 17(a)(ii) is a permissive threshold with an observation-based trigger ("becomes apparent").
- Rule 17(b) is a mandatory threshold with a prediction-based trigger (collision cannot be avoided by the give-way vessel alone).
- Citing *The Maloja II*, the Court emphasised that avoiding manoeuvres should be made when the officer is not forced into a quick decision or one on inadequate information, and should be readily apparent to the other ship.

The observation/prediction distinction and its algorithmic consequences (Section 3.4.3) are this paper's reading of the judgment's reasoning, not a statement made in those terms by the Court.

4.6 Operational consequences

1. Do not make course estimation a precondition for engaging the crossing rules. Use bearing rate of change.
2. Keep applying the crossing rules to slow, erratic, manoeuvring and waiting vessels until the entrance test is met.
3. Switch rule sets at a channel entrance on observable evidence of shaping to enter on final approach, not on an inference of intent. A premature switch is as much an error as a late one.
4. Implement the 17(a)(ii) and 17(b) triggers as separate functions with different accuracy requirements, and declare both.
5. Set decision timing so that manoeuvres are not left until they cannot be readily apparent.
6. Declare in the ConOps which jurisdiction's interpretation the logic implements.

4.7 Limits

- The tests are UK law. Their uptake elsewhere is unsurveyed.
- The judgment decided a specific collision. Its translation into sensor thresholds is engineering judgment, not law, and no numeric values come from the court.
- No flag-State circular, class rule or IMO guidance providing an equivalent trigger was identified in the sources used, but practice has not been surveyed.

5. Operational and Design Implications

5.1 ConOps content

The ConOps (MASS Code 8.2, Annex 2) is where most of the declarations recommended in Section 3 should live. At a minimum it should address:

1. Sensor suite, coverage, degradation modes and look-out equivalence (Rule 5).
2. Operational envelope parameters mapped to Rule 6 factors, with degraded-state buffers before the envelope limit.
3. Sensor sufficiency and confidence floor, and the resulting alerts and degraded states (Rule 7).
4. Restricted-visibility mode: definition, speed policy, prohibited manoeuvres, automated sound signals (Rules 19 and 35).
5. Maximum decision latency for avoidance action (Rule 8) and maximum intervention latency from danger detection to master action (Rule 2), by operating phase.
6. Separate thresholds for the Rule 17(a)(ii) observation trigger and the Rule 17(b) prediction trigger, with escalation criteria.
7. Crossing-rule engagement on steady bearing, with the bearing measurement accuracy relied on.
8. Rule set applied by voyage phase and geography, behaviour in narrow channels and traffic separation schemes, and the jurisdictional interpretation of the Rule 9 and crossing-rule interaction.
9. Channel-entrance rule-switching logic, where used (Section 5.4).
10. ANS fallback behaviour, authority transfer and the conditions requiring the master on board.
11. Responsibility-allocation matrix (Section 5.5).

12. Person-presence modes and their effect on MARPOL Annexes IV and V, and the rule for the master's location.
13. ROC description, ROC security, remote-operator qualifications, rest and workload limits.
14. Incident-reporting assignment.

5.2 Approval basis and regulatory gap analysis as the interpretation layer

The mechanism. The Code builds an approval process (Ch. 6) around an approval basis submitted by the submitter and approved by the Administration (Annex 1, 1.4 and 2.5). The ConOps must identify "possible deviations from the applicable rules and regulations" and justify alternative solutions on the basis of risk assessment (Annex 2, 8.2), and the MASS Record must contain a regulatory gap analysis (5.1.3.4). Taken together, these documents give a vessel-specific, Administration-approved reading of how unchanged legacy rules apply to that vessel. Changes to the ConOps, operational envelope, software or ROC require reassessment (6.4.2, 8.2.5).

What it is not. It is not an IMO interpretation, which requires committee action. It is not a general flag-State circular, since it is specific to one vessel. It is not an exemption or equivalent arrangement under SOLAS I/5, which is a different mechanism (operative text to be verified). It is not a court ruling, although it may encode court rulings, as with *Ever Smart*.

Legal force. During the transition its force derives from the Administration's approval and national implementation, not from the non-mandatory Code alone. Where an entry interprets COLREG text, it is an interpretation within the text. Where an entry would depart from the text, the departure has to be justified through Annex 2, 8.2, and remains subject to the mandatory legacy text and the flag State's own authority to accept it.

Why it matters now. Because COLREG is unlikely to change before 2032 (Section 6), this layer is the main route by which COLREG ambiguities are resolved for autonomous operation in the meantime. Submitters that record interpretations clearly and justify them well may influence developing practice.

Content of entries. For each friction in Section 3 rated "open" or "partly addressed", the approval basis should contain a corresponding entry. The principal ones are summarised below.

Topic	What the entry should declare	Where it should sit
Look-out (Rule 5)	Sensor-to-function mapping, degradation modes, equivalence argument under 6.2	Approval basis; ConOps
Safe speed (Rule 6)	Rule 6 factors mapped to envelope parameters and degraded-state buffers	Approval basis; ConOps
Scanty information (Rule 7)	Sensor sufficiency and confidence floor; uncertainty handling	Approval basis; V&V plan
Readily apparent action (Rule 8)	Manoeuvre magnitude and timing criteria; decision latency; MASS-to-MASS signalling	Approval basis; V&V plan
Stand-on triggers (Rule 17)	Separate observation and prediction thresholds; escalation	Approval basis; ConOps; V&V plan
Crossing engagement	Bearing-based trigger and required measurement accuracy	Approval basis; V&V plan
Channel entrances	Jurisdictional interpretation; rule-switching logic if used	ConOps; gap analysis
Signals (Parts C, D)	Activation mechanism; survey checklist	Survey checklist; V&V plan
Remote master (Rule 2)	Intervention latency; fallback response; authority transfer	ConOps
Crew limitations (Rule 2)	Human-system limitation analysis	Gap analysis
Responsibility	Allocation matrix; no-gap verification	ConOps; approval basis

Topic	What the entry should declare	Where it should sit
ISM and ROC	Interfaces; ROC safety management; alternative certification if used	SMS; approval basis
STCW and rest	ROC watchkeeping standard; qualification equivalence	ConOps; approval basis
SOLAS V/19	How requirements are met electronically	Approval basis; ConOps
Stability, fire, life-saving	Human or remote response chains; crewless arrangements	ConOps
MARPOL	Annex I checklist; person-presence mode plan; reporting assignment	ConOps; MASS Record
Security and cyber	ROC in ship security plan; cyber link to COLREG-critical functions	Ship security plan; ConOps
Application and version	Application-gate decision; rule version lock; amendment horizon	MASS Record; approval basis

5.3 Sensor and decision-logic requirements

1. **Bearing-based engagement.** Crossing rules engage on bearing rate of change. Bearing measurement accuracy over a declared time window is the baseline sensing requirement, and course estimation should not be a precondition.
2. **Sensor sufficiency.** A declared confidence floor, below which the ANS alerts and degrades rather than acting on scanty information.
3. **Separate triggers.** An observation-based change detector for the give-way vessel's behaviour (17(a)(ii)) and a prediction model of her manoeuvring capability (17(b)), each with declared accuracy and uncertainty handling.
4. **Early decision.** Decision thresholds set early enough that the manoeuvre is positive, in ample time and readily apparent, with declared maximum decision latency.
5. **Degraded states.** Defined behaviour for loss or degradation of individual sensors, with graduated mode changes before the envelope limit.
6. **Restricted visibility.** A defined mode with its own sensor assumptions, speed policy, prohibited manoeuvres and automated signals.
7. **Rule-set logic.** An explicit state model for rule-set selection at channel entrances and demarcation lines, with declared transition criteria.
8. **Non-cooperative targets.** Declared capability limits and escalation.
9. **Override.** Human override and master intervention available in every mode (8.7.1, 16.5), with latency declared.

5.4 Rule-switching at channel entrances: a proposed design approach

Status. This is a proposal. It is grounded in the *Ever Smart* holding on shaping to enter (Section 4.3) but has not been validated by any flag State, classification society or IMO body. The numeric parameters are engineering proposals and are not derived from the judgment. It should be tested against the consolidated Rule 9 text and the full reasoning of the judgment before use.

States.

State	Name	Definition	Governing rules
S1	Crossing	Approaching vessel not intending to enter, or intending but not yet shaping	Crossing rules

State	Name	Definition	Governing rules
S2	Transition	Approaching vessel adjusting course toward the entrance but not yet committed to final approach	Crossing rules, with enhanced monitoring
S3	Shaping	Approaching vessel shaping to enter on final approach	Rule 9 overrides the crossing rules

S2 is an engineering buffer added by this proposal. It is not a category in the judgment. Group 1 maps to S1. Group 3 maps to S1, or to S2 once a converging course adjustment begins. Group 2 maps to S3.

Observable conditions. Each approaching vessel is assessed on three conditions:

Condition	Question	Typical inputs
C1 Course adjustment	Is she adjusting her course?	Bearing rate, heading rate, course-over-ground change over a time window
C2 Target geometry	Is the adjustment directed at the entrance on her starboard side?	Relative position to entrance geometry from the electronic chart; projected course against the entrance
C3 Final approach	Is she on final approach, rather than manoeuvring, waiting or transiting?	Speed, distance to entrance, closure rate, absence of station-keeping, pilot status where available

Transition logic.

- S1 to S2 when C1 is true and C2 is partial (adjustment converges on the entrance but is not yet aligned).
- S2 to S3 when C1 is sustained, C2 is true and C3 is true.
- S3 to S2 if C3 becomes false (final approach aborted, holding or manoeuvring away).
- S2 to S1 if C1 or C2 becomes false.

Known gaps in the logic as drafted.

1. An approaching vessel already aligned with the entrance and closing, but showing no detectable course adjustment, has no path out of S1. Whether she is shaping to enter is a question the approval basis must answer.
2. There is no direct path from S1 to S3, so a vessel that becomes aligned and closing within one update cycle must pass through S2.
3. Hysteresis is applied to reversions only. Forward transitions rely on the detection window alone.

Proposed default parameters (500 GT and above; vessel-specific values to be declared).

Parameter	Proposed default
C1 detection window	Sustained course change of at least 5° for at least 60 seconds
C2 alignment tolerance	Projected course within ±10° of the entrance axis
C3 closure	Distance to entrance decreasing at 0.5 knots or more and within 2 nautical miles
C3 station-keeping	Speed over ground 1 knot or less for at least 5 minutes, or heading oscillation of ±15° without net closure
Reversion hysteresis	State persists at least 30 seconds before reversion

Proposed sensor performance. Radar bearing accuracy of 1° or better and range accuracy of 1% of range or better; AIS position accuracy of 10 m or better with updates every 2 seconds or faster; GNSS accuracy of 5 m or better; electro-optical and infrared sensing sufficient to identify navigation lights at 2 nautical miles or more. Chart accuracy should be specified by reference to the applicable electronic chart standard. An earlier draft cited IHO S-44, which, as far as is known, is the hydrographic survey standard rather than an electronic chart product specification, so the intended reference should be confirmed. Without radar, AIS with electro-optical sensing may support C1 and C2 but not closure rate. If both radar and AIS are lost, the ANS should enter a degraded state and

alert the master.

ANS actions by state (proposed).

State	Action	Notification
S1	Apply crossing rules: early and substantial action if give-way, maintain course and speed and monitor for the Rule 17 triggers if stand-on	Standard watchkeeping alerts
S2	Apply crossing rules, increase bearing observation frequency, prepare for a possible change	Advisory that the approaching vessel may be shaping to enter
S3	Apply narrow-channel rules, including Rule 9(a), and cease treating the situation as a crossing	Alarm that the governing rule set has changed

The S3 behaviour, including the position of a MASS that is itself the inbound or outbound vessel, should be checked against the Rule 9 text and the judgment's reasoning before adoption.

Verification tests to include in the V&V plan.

Test	Scenario	Pass criterion
V1	Vessel crossing the entrance, not intending to enter	Remains in S1 throughout
V2	Vessel shaping to enter on final approach	S1 to S2 to S3 at the correct point
V3	Vessel waiting outside the entrance for a pilot	Remains in S1 or S2, with no premature switch to S3
V4	Vessel begins shaping, then aborts	Correct reversion
V5	Erratic-course give-way vessel	Crossing rules engage on steady bearing despite unsteady course
V6	Radar lost	Degraded state and alert; no transition on insufficient data
V7	Several vessels at the entrance	Correct state per vessel, with no cross-contamination
V8	Night operation, lights only	State logic works without radar using electro-optical confirmation

A test for the first known gap above (an aligned, closing vessel with no detectable course adjustment) should be added.

5.5 Responsibility allocation

The matrix below is a starting point for the ConOps. The right-hand column shows where the Code speaks and where the allocation is left open.

Function	Administration	Company	ROC	Master	ANS	Code position
Application-gate decision; approval of the approval basis and ConOps	Decides and approves	Submits (or via submitter)	Described in submission	Not a decision-maker	Not applicable	Ch. 2, 6
Definition and maintenance of the operational envelope	Approves	Defines and maintains	Operates within	Informed; may intervene on exit	Operates within	8.3–8.5
Safe operation of the MASS at all times	Oversees through certification	Provides management system and resources	Provides monitoring and control interface	Responsible	Acts as agent; not a master	8.7.3, 14.2.1.10
COLREG-compliant navigation	Certifies the approach	Ensures the system and procedures	Supervises	Responsible	Executes	16.2, 16.4

Function	Administration	Company	ROC	Master	ANS	Code position
Intervention and override	Verifies capability	Provides means and procedures	Provides interface	Exercises	Yields on override	8.7.1, 16.5; latency left to ConOps
Fallback activation and authority	Approves arrangements	Defines in SMS	Monitors triggers	Directs	Executes pre-programmed fallback	8.4; interval left to ConOps
ROC management and certification	Issues ROC certificates	Operates the ROC	Subject of certificates	Not applicable	Not applicable	5.3, 11
Security (ship security plan including ROC)	Approves plan	Prepares and maintains	Takes part in drills; isolation on compromise	Not applicable	Fallback on breach	12.4
Qualification, rest and workload of remote operators	Accepts equivalence	Implements standard	Applies	Not applicable	Not applicable	14; quantification left to ConOps
Incident and pollution reporting	Receives	To be assigned	To be assigned	To be assigned	Detects and logs	9.10.5; reporting actor not specified
Data logging and access	Access on request	Retains	Retains	Not applicable	Logs	9.10

Where the Code is silent on who acts, the ConOps and SMS should say so explicitly, and the approval process should confirm that every function has an assigned human or corporate owner.

5.6 Shared dependencies and ROC concentration

- The Code requires each ROC to hold a valid certificate for each MASS it operates (5.3.1). If a ROC certificate is withdrawn or lapses, the Administration is to review the effect on the associated MASS Safety Certificate and on other ROC certificates for that ROC (5.6.2). A single ROC serving several MASS creates a many-to-one dependency. A ROC failure (cyber compromise, connectivity loss, personnel failure) can affect all associated MASS at once. The Code does not limit the number of MASS per ROC or require redundancy.
- A flag State could consider requiring or encouraging ROC redundancy, or limiting or disclosing the number of MASS that depend on one ROC (a proposal, not a finding). The ConOps should state that number and the fallback arrangements.
- The regulatory gap analysis is a per-vessel document and does not capture fleet-level exposure. An approval package should therefore disclose shared dependencies: the ANS software supply chain, shared ROC, communication, update, key-management and cloud services, vessels sharing a design family, and common COLREG interpretations, Rule 17 thresholds, Rule 2 departure logic and restricted-visibility behaviour. Common interpretations are not a technical vulnerability in themselves, but they mean that one flawed interpretation is replicated across a fleet.
- V&V evidence should cover degraded sensors, calibration drift and environmental drift, and should address the gap between model and reality that the Code's testing and validation provisions (9.12) leave unquantified.

6. Transition Period Considerations (2026–2032)

Voluntary status. From 1 July 2026 a MASS may be approved against a voluntary Code alongside mandatory legacy instruments. Consequences:

- The Code's provisions bind only to the extent a flag State implements them nationally or an owner, charterer or class society requires them. Uptake may be uneven.
- Flag States may reach different conclusions on the same design. Approval certificates and the approval basis should state clearly what was assessed, against which instruments, and on what interpretation.
- The mandatory text remains the standard against which a casualty would be judged. The Code's equivalence principle is therefore best understood as a route to approval, not a defence.

Practical freeze on COLREG amendment. COLREG was last amended in 2013 (as recorded in the Status of Treaties volume), and no further amendment cycle was identified in the sources used. Amendment follows the procedure in Article VI: circulation to Parties at least six months before consideration, adoption by the Maritime Safety Committee, adoption by the Assembly, and entry into force on a date set by the Assembly unless more than one-third of Parties object. That route is slower than SOLAS tacit acceptance. With the roadmap target at 1 January 2032, COLREG text is practically unlikely to change for roughly five years. That is a practical expectation, not a legal bar, and could change if an extraordinary cycle is convened.

Consequences of that freeze:

- Harmonisation between COLREG and autonomous operation has to happen through interpretation, ConOps declarations and approval, not amendment.
- Flag States have until about 2032 to develop interpretive frameworks. Early approvals with robust COLREG interpretations may shape practice elsewhere.
- Case law such as *Ever Smart* carries more weight, and the question of which jurisdiction's interpretation applies becomes a standing item in every approval.

Version management. Amendments to SOLAS and MARPOL continue to be adopted and will enter into force during the transition. Approval bases should be version-locked and amended on defined triggers.

Small and military vessels. MASS below 500 gross tonnage and government non-commercial vessels operate outside the Code's mandatory ambit even after 2032. Civil MASS may meet military autonomous vessels operating under no common framework, and government vessels may claim sovereign immunity. No IMO-level solution exists.

Recommended actions during the transition.

Audience	Suggested action
Submitters	Prepare the approval basis and gap analysis as the main interpretive record. Declare thresholds rather than relying on the Code's general language.
Administrations	Use Section 3 and the Section 5.2 table as a checklist for completeness. Record the application-gate decision and the rule version.
Classification societies	Develop survey and V&V checklists for signalling, automated discharge monitoring, ROC functions and rule-switching logic.
ROC operators and Companies	Incorporate declared thresholds into the SMS. Train operators on them. Declare ROC concentration.
Developers	Design for bearing-based engagement, separate Rule 17 triggers, early decision timing and explicit rule-set state models.

7. Open Issues and Verification Needs

Primary-text verification (required before external reliance).

1. Check all MASS Code paragraph references and the adoption, effect and roadmap dates against the official MSC.595(111) text.
2. Check all operative COLREG wording against the consolidated text as amended. The operative text used here comes from the *Ever Smart* judgment and a secondary source. Rules 1, 10, 20–37 and 39–41 have not been checked even against those.
3. Verify operative SOLAS text (V/19, V/20, V/33, XI-1/6, I/5, I/12), STCW A-VIII/1, the Load Lines Convention and IS Code, and the MARPOL annexes and Protocol I.
4. Re-check all judgment paragraph references (paras 40, 45, 60–61, 111–112, 134–145) against the full judgment.
5. Verify the Status of Treaties data used (amendment dates, the III Code's entry into force, the 2027 and 2028 entry-into-force dates).
6. Verify the national inland rules referred to at 3.5.3 and the other rule-set boundaries that matter to a given vessel.

Legal and jurisdictional scope.

1. Which interpretation of the Rule 9 and crossing-rule interaction flag, coastal and port States outside the UK apply is unsurveyed.
2. Flag-State practice on small MASS and ROC watchkeeping is unsurveyed.
3. Whether the 2032 roadmap date holds, and whether COLREG remains unamended to that date.
4. How surveyors and Administrations will treat unmanned machinery spaces under IOPP surveys, and ROC rest and watchkeeping under STCW.

Design proposals needing validation.

1. The rule-switching state machine, its parameters and sensor figures (Section 5.4) have not been validated by any authority. The three known logic gaps are unresolved.
2. The S3 behaviour for a MASS that is itself inbound or outbound needs checking against the Rule 9 text and the judgment's reasoning.
3. The appropriate electronic chart accuracy reference needs to be confirmed.
4. Acceptable sensor sufficiency, prediction accuracy and latency values have no authoritative source and need work in IMO, class or industry bodies.
5. The cyber-to-COLREG dependency is an inference and should be tested against a real ROC architecture, including actual ROC concentration and shared dependencies.
6. The external precedents for capture of unmanned platforms cited in earlier analysis have not been independently verified here and are not relied on.

Document status.

1. This paper has not been independently reviewed. It was rebuilt from an earlier working draft by a single producing session and should receive review by a reader with access to the primary texts before circulation.

Annex A. Glossary

Term	Meaning
ANS	Autonomous navigation system
ConOps	Concept of operations
CPA / TCPA	Closest point of approach / time to closest point of approach
ECDIS / ENC	Electronic chart display and information system / electronic navigational chart
IOPP	International Oil Pollution Prevention certificate
III Code	IMO Instruments Implementation Code
MASS	Maritime autonomous surface ship
OE	Operational envelope (MASS Code 8.3.1)
ROC	Remote operation centre
SMS	Safety management system
V&V	Verification and validation

Annex B. Principal Sources

- MSC.595(111), International Code of Safety for Maritime Autonomous Surface Ships (MASS Code), including the roadmap annex.
- IMO, *Status of IMO Treaties* (30 October 2026).
- *Evergreen Marine (UK) Ltd v Nautical Challenge Ltd* [2021] UKSC 6.
- COLREG 1972, SOLAS 1974, MARPOL 73/78, STCW, ISM Code, ISPS Code, Load Lines Convention, III Code (operative texts to be verified; see Section 7).

Annex C. Illustrative Entries

These are illustrative drafting aids for the channel-entrance logic in Section 5.4. They are proposals. The statement about validation is true only once the tests have been run.

Illustrative ConOps declaration.

> *This MASS implements a three-state rule-switching model for narrow channel entrances, based on the interpretation in Evergreen Marine (UK) Ltd v Nautical Challenge Ltd [2021] UKSC 6, para 145, as adopted by [flag State / coastal State]. The ANS selects the applicable COLREG rule set from three observable conditions: course adjustment, target geometry and final approach. In state S1 (crossing) and S2 (transition), Rules 15–17 apply, with enhanced monitoring in S2. In state S3 (shaping), Rule 9 applies. The ANS does not switch rule sets for a vessel that is merely intending or preparing to enter, for example one waiting for a pilot. Declared thresholds: [vessel-specific values]. The state model reverts if the approaching vessel aborts final approach. Validation: [results of tests V1–V8 and any additional tests].*

Illustrative regulatory gap analysis entry.

Field	Content
Gap	COLREG Rule 9 and Rules 15–17 interaction at channel entrances
Affected rules	COLREG Rules 9, 15, 16, 17
Nature of gap	The COLREG text does not state the condition for switching between crossing rules and narrow-channel rules at a channel entrance. The UK Supreme Court has stated a test (shaping to enter). It is jurisdiction-specific and not reflected in IMO guidance.
Deviation	The ANS applies the three-state model. Where the Administration regards this as an interpretation within the COLREG text, no deviation arises. Where it is regarded as a departure from a purely textual reading, it is declared as such under Annex 2, 8.2.
Justification	The test is the authoritative, sensor-observable trigger identified for this purpose. It is consistent with the purpose of COLREG and with the evaluation principle at MASS Code 6.2.
Alternatives considered	(1) Apply narrow-channel rules near all entrances: over-broad and inconsistent with the judgment. (2) Apply crossing rules at all times: inconsistent with Rule 9. (3) Flag-State-specific rule: adopted and declared in the ConOps.
Risk assessment reference	[Reference to risk assessment under MASS Code Ch. 7]

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